Instituto Politécnico Nacional

Escuela Superior de Cómputo

Selected Topics in Cryptography

Reporte Práctica 01:

“Non-singular elliptic curves”

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Programming exercises

Design a function that receives as input a prime number p 3 to find the quadratic residues modulo p and also the square roots modulo p. For example if p = 11 and we know that 22 mod 11 = 4 and 92 mod 11 = 4 we know that 4 is a quadratic residue and its square roots are 2 and 9

```python
from sympy import isprime, randprime


def residuos_cuadraticos(p):
    if p < 2 or not isprime(p):
        raise ValueError("El numero debe ser primo y mayor a 2")
    residuos = {}
    for x in range(p):
        residuo = (x * x) % p
        if residuo not in residuos:
            residuos[residuo] = []
        residuos[residuo].append(x)
    return residuos

p = randprime(3,100)
residuo_cuadrados = residuos_cuadraticos(p)
print(f"Residuos cuadraticos y raices cuadradas {p}: {residuo_cuadrados}")
```

```
> & C:/Users/saids/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/saids/Documents/ESCOM/8voSemestre/CRIPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio1.py"
Residuos cuadraticos y raices cuadradas 29: {0: [0], 1: [1, 28], 4: [2, 27], 9: [3, 26], 16: [4, 25], 25: [5, 24], 7: [6, 23], 20: [7, 22], 6: [8, 21], 23: [9, 20], 13: [10, 19], 5: [11, 18], 28: [12, 17], 24: [13, 16], 22: [14, 15]}
PS C:\Users\saids\Documents\ESCOM\8voSemestre\CRIPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY>
```

Design a function that receives as input a prime number p 3, and generates a non-singular elliptic curve over Zp, ie. your function must output the parameters ab such that the discriminant is different from zero.

```python
from sympy import isprime, randprime
import random

def generar_ec_no_singular(p):
    if p < 2 or not isprime(p):
        raise ValueError("El número debe ser primo y mayor a 2")

    while True:
        a = random.randint(1, p - 1)
        b = random.randint(1, p - 1)
        if (4 * pow(a, 3) + 27 * pow(b, 2)) % p != 0:
            return f'y^2 = x^3 + {a}x + {b}'

p = randprime(3, 100)
print(f"Ecuación de la curva para el primo {p}: {generar_ec_no_singular(p)}")

```

Descripción de la función:

Verifica que p sea primo y mayor a 2, posteriormente en el while al azar escoge 2 valores de a y b entre 1 y 1-p tal que al calcular el discriminante este sea distinto de 0. Si se comprueba esto procedemos a armar la ecuación de la curva y la retornamos.

```
Ecuación de la curva para el primo 3: y^2 = x^3 + 2x + 2
PS C:\Users\saids\Documents\ESCOM\8voSemestre\CRIPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY> & C:/Users/saids/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/saids/Documents/ESCOM/8voSemestre/CRIPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY/Practicas/Practica01/Ejercicio2.py"
Ecuación de la curva para el primo 7: y^2 = x^3 + 2x + 6
PS C:\Users\saids\Documents\ESCOM\8voSemestre\CRIPTOGRAFIA SELECTED TOPICS IN CRYPTOGRAPHY>
```